

CH8.2 正餘弦定理

$$1. 2\sqrt{2} \quad 2. \sqrt{19} \quad 3. 7 \quad 4. \sqrt{65} \quad 5. \sqrt{2} \quad 6. \frac{3}{4} \quad 7. \frac{\sqrt{2}}{2} \quad 8. 3\sqrt{3}, 3\sqrt{3}, \frac{9\sqrt{3}}{2} \quad 9. \frac{7}{10}$$

解析

1. 令此圓之半徑為 R

$$\text{在 } \triangle ACD \text{ 中, 由正弦定理知 } \frac{2}{\sin 30^\circ} = 2R \cdots \textcircled{1}$$

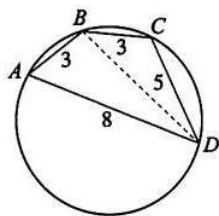
$$\text{在 } \triangle ABC \text{ 中, } \frac{\overline{AB}}{\sin 45^\circ} = 2R \cdots \textcircled{2}$$

$$\text{由 } \textcircled{1}\textcircled{2} \text{ 知 } \frac{2}{\sin 30^\circ} = \frac{\overline{AB}}{\sin 45^\circ} \Rightarrow \overline{AB} = \frac{2\sin 45^\circ}{\sin 30^\circ} = \frac{2 \times \frac{\sqrt{2}}{2}}{\frac{1}{2}} = \underline{2\sqrt{2}}$$

$$2. \triangle ABC \text{ 中, 餘弦定理得 } \cos B = \frac{49+25-9}{2 \cdot 7 \cdot 5} = \frac{13}{14}$$

$$\triangle ABD \text{ 中, } \overline{AD}^2 = 3^2 + 7^2 - 2 \cdot 7 \cdot 3 \cdot \cos B = 9 + 49 - 2 \cdot 7 \cdot 3 \times \frac{13}{14} = 58 -$$

$$39 = 19 \Leftrightarrow \overline{AD} = \sqrt{19}$$



3. $\because \angle BCD$ 與 $\angle BAD$ 互補

$$\therefore \cos C = -\cos A$$

設 $\overline{BD}$ 長 x

$$\therefore \frac{64 + 9 - x^2}{48} = \frac{x^2 - 34}{30}$$

$$\Rightarrow 365 - 5x^2 = 8x^2 - 272 \Rightarrow 13x^2 = 637$$

$$\Rightarrow x^2 = 49 \Rightarrow x = \underline{7}$$

4. 在直角 $\triangle ABC$ 中, $\sin C = \frac{3}{5}$

$$\begin{aligned} \text{在 } \triangle ACD \text{ 中 } \overline{AD}^2 &= \overline{AC}^2 + \overline{CD}^2 - 2 \times \overline{AC} \times \overline{CD} \times \cos \angle ACD = 16 + 25 - \\ 2 \times 4 \times 5 \times \cos(90^\circ + \angle ACB) &= 41 - 40 \times (-\sin \angle ACB) = 41 - 40 \times \left(-\frac{3}{5}\right) = \\ 65 \therefore \overline{AD} &= \sqrt{65} \end{aligned}$$

5. $\because \angle ADB = \angle ACB = 30^\circ$

$$\text{令此圓之半徑為 } R, \text{ 則 } \frac{\overline{AB}}{\sin \angle ACB} = 2R = \frac{\overline{AD}}{\sin \angle ACD}$$

$$\Leftrightarrow \frac{\overline{AB}}{\sin 30^\circ} = \frac{2}{\sin 45^\circ} \Leftrightarrow \overline{AB} = \frac{2}{\frac{1}{\sqrt{2}}} \times \frac{1}{2} = \frac{\sqrt{2}}{2}$$

6. $\because \overline{AD}$ 為 $\angle BAC$ 之角平分線

$$\therefore \overrightarrow{AB} : \overrightarrow{AC} = \overrightarrow{BD} : \overrightarrow{DC} = 3 : 6 = 1 : 2$$

令 $\overrightarrow{AB} = \overrightarrow{AD} = x$, 則 $\overrightarrow{AC} = 2x$

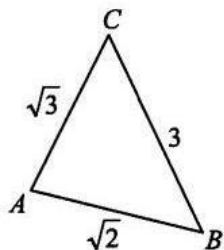
$$\because \cos \angle ADC = \cos(180^\circ - \angle ADB) = -\cos \angle ADB$$

$$\begin{aligned} \therefore \frac{\overline{AD}^2 + \overline{DC}^2 - \overline{AC}^2}{2 \times \overline{AD} \times \overline{DC}} &= -\frac{\overline{AD}^2 + \overline{DB}^2 - \overline{AB}^2}{2 \times \overline{AD} \times \overline{DB}} \\ \Rightarrow \frac{x^2 + 36 - 4x^2}{2 \times x \times 6} &= -\frac{x^2 + 9 - x^2}{2 \times x \times 3} \Rightarrow 36 - 3x^2 = -18 \Rightarrow x^2 = 18 \end{aligned}$$

$$\therefore x = 3\sqrt{2}$$

$$\begin{aligned} \therefore \cos \angle BAD &= \frac{\overline{AB}^2 + \overline{AD}^2 - \overline{BD}^2}{2 \times \overline{AB} \times \overline{AD}} = \frac{18 + 18 - 9}{2 \times 3\sqrt{2} \times 3\sqrt{2}} \\ &= \frac{27}{36} = \frac{3}{4} \end{aligned}$$

$$7. \cos A = \frac{(\sqrt{3})^2 + (\sqrt{2})^2 - 3^2}{2 \cdot \sqrt{3} \cdot \sqrt{2}} = \frac{-4}{2\sqrt{6}} = \frac{-2}{\sqrt{6}}$$



$$\sin A = \sqrt{1 - \cos^2 A} = \sqrt{\frac{1}{3}}$$

$$\therefore \triangle ABC \text{ 面積} = \frac{1}{2} \cdot \sqrt{3} \cdot \sqrt{2} \cdot \frac{1}{\sqrt{3}}$$

$$= \frac{\sqrt{2}}{2}$$

$$8. (1) \triangle ABD = \frac{1}{2} \cdot \overline{AB} \cdot \overline{BD} \cdot \sin 30^\circ = \frac{1}{2} \cdot 6 \cdot 2\sqrt{3} \cdot \frac{1}{2} = \underline{3\sqrt{3}}$$

(2) 設 $\overline{BC} = x$ ，則由 $\triangle ABD + \triangle BDC = \triangle ABC$ 得

$$\frac{1}{2} \cdot 6 \cdot 2\sqrt{3} \cdot \sin 30^\circ + \frac{1}{2} \cdot 2\sqrt{3} \cdot x \cdot \sin 30^\circ$$

$$= \frac{1}{2} \cdot 6 \cdot x \cdot \sin 60^\circ$$

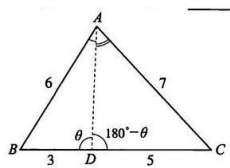
$$\therefore 6\sqrt{3} + \sqrt{3}x = 3\sqrt{3}x \quad \therefore 6\sqrt{3} = 2\sqrt{3}x \quad \therefore x = 3$$

於 $\triangle ABC$ 中, 利用餘弦定律得

$$b^2 = c^2 + a^2 - 2ac \cos B = 6^2 + 3^2 - 2 \cdot 6 \cdot 3 \cdot \cos 60^\circ = 27$$

$$\therefore b = 3\sqrt{3} \quad \therefore \overline{AC} = \underline{3\sqrt{3}}$$

$$(3) \triangle ABC = \frac{1}{2} \cdot \overline{AB} \cdot \overline{BC} \cdot \sin 60^\circ = \frac{1}{2} \cdot 6 \cdot 3 \cdot \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{2}$$



$$9. \frac{\sin \angle BAD}{3} = \frac{\sin \theta}{6} \dots\dots \textcircled{1}$$

$$\frac{\sin \angle DAC}{5} = \frac{\sin(180^\circ - \theta)}{7}$$

$$\frac{\sin \angle DAC}{5} = \frac{\sin \theta}{7} \dots\dots \textcircled{2}$$

$$\frac{\textcircled{1}}{\textcircled{2}} \text{ 得 } \frac{\sin \angle BAD}{3} \div \frac{\sin \angle DAC}{5} = \frac{1}{6} \div \frac{1}{7}$$

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{5}{3} = \frac{7}{6}$$

$$\frac{\sin \angle BAD}{\sin \angle DAC} = \frac{7}{6} \times \frac{3}{5} = \frac{7}{10}$$